CPTP-Invariant Obstruction Theory: Trace-Norm Geometry and Quantum Dual Witnesses

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May 2026

Version: v1.0 (Pre-Print)

Abstract

Within the obstruction framework, the M-compatible gauge classification maps algebraic or spatial symmetry constraints onto induced measurement geometries. For classical scalar networks under maximal bounded-linear contraction, the classification yields uniform ℓ^1 evaluation (Paper 000). For architectures with rotational symmetry $\mathrm{SO}(k)$, the metric reduces to the mixed Euclidean geometry $\ell^1(E;\ell^2)$ (Paper 001b).

This paper extends the classification to open quantum systems. When a finite directed graph transports quantum states bounded by Completely Positive Trace-Preserving (CPTP) evolutions, CPTP monotonicity restricts the admissible gauges to CPTP-monotone distinguishability measures. Combined with tensor stability, classical total-variation reduction, and binary discrimination axioms, these operational assumptions select the trace norm as the local gauge. Under replica-extensive edge-additive aggregation, this yields $\ell^1(E;\text{trace norm})$ for that observer class.

We establish the Φ_{tr} quotient obstruction bound and construct its dual. Once the trace-distance gauge is chosen, the maximal dual witnesses are divergence-free observable circulations bounded by the operator-norm ball. Because this paper defines $\|X\|_{\mathrm{tr}} = \frac{1}{2}\|X\|_1$, the dual bound is $\|M_e\|_{\mathrm{op}} \leq \frac{1}{2}$; using the unhalved trace norm would instead give $\|M_e\|_{\mathrm{op}} \leq 1$. The uniqueness of the trace norm is conditional on the operational axioms listed in Theorem 2.3 and does not follow from CPTP monotonicity alone.

Note: This paper identifies $\ell^1(E;\text{trace norm})$ as the obstruction geometry for replica-extensive observers whose local gauge is the operational trace distance. CPTP monotonicity is necessary for this choice but not sufficient to make it unique; uniqueness requires additional operational assumptions such as classical total-variation reduction and binary discrimination duality.

1 Introduction and Structural Placement

This paper is the CPTP-symmetry instance of the M-compatible gauge classification program (Paper 001). It maps how CPTP-symmetry constraints, together with operational axioms, specify trace-distance geometry inside the replica-extensive ℓ^1 observer framework. It does not propose a universal quantum metric.

The rigorous categorical classification of diagnostic geometries formalizes that structural obstruction constraints intrinsically adapt upon mapping across distinct physical regimes. The universal baseline scalar mapping (Paper 000) derives precisely from requiring full maximal contraction mapping independently over decoupled fields (Paper 001). Vector spaces experiencing rotationally symmetric bindings naturally compress evaluations algebraically to Euclidean invariants (Paper 001b).

Extending this geometry to evaluating quantum mechanical network fields demands mapping local topological discrepancy data onto the state space of density matrices. In this context, nodes represent isolated quantum sub-systems or spatial points, and directed graph connections parameterize expected causal or spatial transport channels. A physical network evaluating quantum failures must specify which operational distinguishability gauge it uses; CPTP monotonicity alone does not make that choice unique.

This imposes CPTP contraction as a necessary compatibility condition. The trace norm becomes the selected gauge only after the additional operational assumptions in Theorem 2.3 are imposed.

2 CPTP-Compatible Gauges and Trace Norm Extraction

Formalizing the observer triple $\mathcal{O} = (\mathcal{A}, \mathcal{M}, \mathcal{J})$, we fix the aggregation law $\mathcal{A}$ to satisfy Replica Extensivity plus edge decomposability, giving linear edge summation. The local geometry then depends on the chosen CPTP-compatible distinguishability gauge.

2.1 The Carrier Stalk and Defect Tensor

All Hilbert spaces discussed are strictly finite-dimensional. Let the local state space at each node v be bounded mathematically by $\mathcal{H}_v$. The state transported is a density operator $\rho_v \in D(\mathcal{H}_v)$. Because network differences subtract states structurally, the local transport defect along a directed edge $e = (a \to b)$ exists functionally as a Hermitian trace-zero operator difference:

$$\delta_e(s) = \Lambda_{b,a}(\rho_b) - \rho_a \in B(\mathcal{H})_0 \tag{1}$$

where $\Lambda_{b,a}$ is the structural pull-back or expected system transition bound.

2.2 CPTP Invariance and Distinguishability Monotones

We define $\|X\|_{\mathrm{tr}} := \frac{1}{2}\|X\|_1$ so that the gauge coincides exactly with quantum trace distance throughout this document.

The observer's metric classification must survive physical application. Specifically, the local gauge $j(\delta_e)$ must contract physically tracking solely under application of any physical open-system channel.

Therefore, the restriction mapping bounding the geometry requires that for any CPTP map $\mathcal{E}$:

$$j(\mathcal{E}(\delta)) \leq j(\delta) \tag{2}$$

By the M-compatible convex body classification (Paper 001, Section 3.6), the gauges retaining monotonic consistency under the CPTP monoid are the **quantum distinguishability monotones**.

2.3 Eliminating Geometric Degeneracy (The Conditional Trace Endpoint)

Theorem 2.3 — Conditional Trace-Norm Rigidity for CPTP-Compatible Gauges

Let V_n be the real vector space of Hermitian trace-zero operators on a finite-dimensional Hilbert space $\mathcal{H}_n$. Let $j_n : V_n \to \mathbb{R}_{\geq 0}$ be a family of gauges satisfying:

1. **Norm structure:** j_n is faithful, convex, homogeneous, and subadditive.
2. **CPTP monotonicity:** $j_m(\Lambda X) \leq j_n(X)$ for every CPTP map $\Lambda : B(\mathcal{H}_n) \to B(\mathcal{H}_m)$.
3. **Unitary invariance:** $j_n(UXU^\dagger) = j_n(X)$.

4. **Tensor stability:** $j_{nk}(X \otimes \rho) = j_n(X)$ for every trace-one density matrix ρ.
5. **Classical reduction:** For any diagonal trace-zero $X = \mathrm{diag}(p - q)$, $j_n(X) = \frac{1}{2}\|p - q\|_1$.

Then $j_n(X) = \frac{1}{2}\|X\|_1$ for every Hermitian trace-zero X.

Proof Skeleton:

1. Unitary invariance implies j_n depends only on the spectrum of X. Therefore $j_n(X) = g_n(\lambda(X))$ for a symmetric gauge g_n on trace-zero eigenvalue vectors.
2. CPTP monotonicity implies contractivity under dephasing: $j_n(\mathcal{D}_U(X)) \leq j_n(X)$. Thus the diagonal classical shadow of X cannot have a larger gauge evaluation than X.
3. Classical reduction fixes the gauge on all diagonal trace-zero operators: $j_n(\mathrm{diag}(x)) = \frac{1}{2}\|x\|_1$.
4. By spectral decomposition, $X = U\mathrm{diag}(\lambda(X))U^\dagger$. By unitary invariance, $j_n(X) = j_n(\mathrm{diag}(\lambda(X)))$, reducing to the diagonal case.
5. Classical reduction then forces $j_n(X) = \frac{1}{2}\|\lambda(X)\|_1 = \frac{1}{2}\|X\|_1$. ■

Note on tensor stability: Tensor stability is included as part of the operational observer package, but the displayed finite-dimensional uniqueness argument uses primarily unitary invariance and exact classical reduction on diagonal trace-zero operators. Tensor stability ensures the gauge family is coherent across composite systems but is not invoked in the spectral reduction steps above.

Trace-norm uniqueness is therefore not a consequence of CPTP monotonicity alone. CPTP monotonicity restricts admissible gauges to quantum distinguishability monotones. The trace norm is isolated only after imposing the additional operational boundary that diagonal quantum defects reduce exactly to classical total variation, together with unitary invariance, tensor stability/classical extension behavior, and norm structure. Under these assumptions, every Hermitian trace-zero defect is unitarily equivalent to a diagonal classical defect with the same spectrum, and the gauge must equal one half of the spectral ℓ^1 norm, i.e., the trace distance.

3 The Global $\ell^1(E; \mathbf{trace})$ Geometry

Given the trace norm isolated under the stated operational axioms, we apply the global spatial constraints structuring the full macroscopic graph independently.

Under Replica Extensivity with edge decomposability, the global diagnostic is the linear edge summation of local gauges:

$$J_{\mathrm{tr}}(\delta) = \sum_e \|\delta_e\|_{\mathrm{tr}} \tag{3}$$

Thus, integrating the trace-distance local gauge with a replica-extensive, edge-decomposable network observer yields the structural state-space metric $\ell^1(E; \mathrm{trace\ norm})$.

4 Quantum Quotient Obstruction and Dual Observables

4.1 The Primal Form

The quotient obstruction is the infimum of the global diagnostic over all coboundary corrections:

$$\Phi_{\mathrm{tr}}([\delta]) = \inf_{\alpha \in C^0} \sum_e \|\delta_e - (d_0\alpha)_e\|_{\mathrm{tr}} \tag{4}$$

where $\alpha \in C^0$ assigns a Hermitian trace-zero correction $\alpha_v \in B(\mathcal{H})_0$ to each vertex v, and $(d_0\alpha)_e = \alpha_{t(e)} - \alpha_{s(e)}$ is the coboundary evaluated at edge e.

4.2 Quantum Dual Witness Theorem

The Banach dual of the trace norm is the operator norm. The conjugate identity is:

$$\left(\frac{1}{2}\|\cdot\|_1\right)^* = \iota_{\{\|M\|_{\mathrm{op}} \leq 1/2\}} \tag{5}$$

where ι is the indicator function of the operator-norm ball. Minimization over coboundary corrections α in the primal imposes the divergence-free constraint $d_0^* M = 0$ on the dual variables.

Theorem 4.1 (Quantum Dual Witness Formulation).
Let P be a finite directed graph. The quantum quotient obstruction satisfies:

$$\Phi_{\mathrm{tr}}([\delta]) = \max_M \sum_e \mathrm{Tr}(M_e \delta_e) \tag{6}$$

subject to:

1. **Divergence-Free:** $d_0^* M = 0$, i.e. $\sum_{\mathrm{in}} M_e - \sum_{\mathrm{out}} M_e = 0$ at every vertex.
2. **Dual Ball:** $\|M_e\|_{\mathrm{op}} \leq \frac{1}{2}$, conjugate to the primal gauge $\|X\|_{\mathrm{tr}} = \frac{1}{2}\|X\|_1$. Under the unhalved trace norm, the dual bound would be $\|M_e\|_{\mathrm{op}} \leq 1$.

5 Discussion: The Observable Loop Failure

The duality relationship structurally converts abstract boundary calculations into explicit physical experiment parameterizations. The functional elements M_e map conceptually onto macroscopic network observables. To explicitly measure the inherent topological contradiction residing inside the global network bounds without local gauge dependence:

- We seek a system of physically deployable observables ($\|M_e\|_{\mathrm{op}} \leq \frac{1}{2}$ under the trace-distance convention).
- We demand that these observables satisfy flow conservation (divergence-free logic over operators).
- The summation of their local expectation values evaluates the invariant graph discrepancy continuously, maximizing to precisely capture the unique topological boundary bounds.

This mathematically restricts the topological bounds tracking directly onto experimentally available observable traces.

6 Conclusion

By incorporating CPTP compatibility into the gauge classification, we obtain a conditional extension of obstruction calculus to open quantum systems. Under the operational axioms of Theorem 2.3, the Φ_{tr} quotient obstruction and its divergence-free operator-norm-bounded dual witnesses are well-defined on finite directed graphs with finite-dimensional Hermitian trace-zero edge defects.

Field	Value
Status	Quantum observer-geometry instance
Carrier	Finite directed graph with finite-dimensional Hermitian trace-zero edge defects
Observer	Replica-extensive edge aggregation + trace-distance local gauge
Main claim	Trace-distance local gauge + edge-additivity yields $\ell^1(E;\text{trace norm})$
Non-claims	CPTP monotonicity alone does not force trace norm; no derivation of quantum mechanics; no Born-rule derivation; no universal quantum metric claim

Status and Non-Claims

References

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